Notes

Introduction to Graphs in Physics

Overview

In this session, we will continue our discussion on uniformly accelerated motion, focusing specifically on the application and interpretation of graphs. Graphs are a very important topic for the JEE exam, and we will explore various questions related to them.

Previous Session Recap

In the last session, we covered numerous questions related to vertical gravity and uniformly accelerated motion. Today's session builds upon those concepts by introducing graphical analysis.

Introduction to Graphs and Calculus Basics

Recalling Graph Concepts

We will quickly recall how to use graphs, especially in the context of calculus, which is crucial for physics. While you might have studied "Area Under Curve" in Class 12 mathematics (involving integrals like $\int y\,dx$), we will focus on the basics and their application in physics.

Understanding dy/dx

If y is a function of x, meaning y depends on x, we can represent this relationship graphically with y on the vertical axis and x on the horizontal axis.

- **Definition:** $\frac{dy}{dx}$ represents the **slope of the tangent** to the curve at a given point. It is also referred to as the gradient.
- **Graphical Interpretation:** If we have a graph of y versus x, and we want to find $\frac{dy}{dx}$ at a specific point P, we draw a tangent line to the curve at that point. The slope of this tangent line gives the value of $\frac{dy}{dx}$ at point P.

Understanding Integral y dx

- **Definition:** The integral $\int y\,dx$ gives the **area under the graph** of y versus x.
- **Graphical Interpretation:** If we integrate y with respect to x from a point x_1 to x_2, the value of the integral corresponds to the area enclosed by the curve, the x-axis, and the vertical lines at x_1 and x_2.
- **Sign Convention for Area in Physics:**
  - **Positive Area:** In physics, any area above the x-axis is considered positive.
  - **Negative Area:** Any area below the x-axis is considered negative. This differs from some mathematical conventions (like clockwise/anticlockwise area) where the sign might indicate

Introduction to Position-Time and Displacement-Time Graphs

Importance of Graphs in Physics

Graphs are crucial tools in physics, not only for the chapter on motion but also for many subsequent topics. They provide a visual representation of physical quantities and their relationships.

Position-Time and Displacement-Time Graphs

These graphs plot position or displacement on the y-axis against time on the x-axis.

- **Y-axis:** Position or Displacement
- **X-axis:** Time

Slope of Position-Time / Displacement-Time Graph

The slope of a graph is generally given by dy/dx. In the case of a position-time or displacement-time graph:

- If the y-axis represents displacement (s), the slope is ds/dt.
- If the y-axis represents position (x), the slope is dx/dt.

Both ds/dt and dx/dt represent **velocity**.

$$\text{Velocity} = \frac{\text{Change in Displacement}}{\text{Change in Time}} = \frac{ds}{dt}$$

$$\text{Velocity} = \frac{\text{Change in Position}}{\text{Change in Time}} = \frac{dx}{dt}$$

Therefore, the **slope of a position-time graph or a displacement-time graph gives the velocity** of the object.

Area Under Position-Time / Displacement-Time Graph

The area under a position-time or displacement-time graph **does not yield any meaningful physical quantity** for our analysis. Therefore, it is not useful for calculations in this context.

Velocity-Time (VT) Graph

Introduction

The velocity-time graph is a crucial tool in kinematics, providing comprehensive information about an object's motion. It plots velocity on the y-axis and time on the x-axis.

Slope of the VT Graph

- Slope = $\frac{dy}{dx}$ = $\frac{\text{Velocity}}{\text{Time}}$ = $\frac{dv}{dt}$
- Since dv/dt is the definition of instantaneous acceleration, the slope of the VT graph directly gives the acceleration of the object.

Area Under the VT Graph

- The area under a velocity-time graph represents displacement.
- The area is calculated by integrating velocity with respect to time.
- Area = $\int y\,dx = \int v\,dt$
- We know that velocity (v) can be expressed as the rate of change of displacement (s) with respect to time (t): $v = ds/dt$.
- Substituting this into the integral: $\int v\,dt = \int \left(\frac{ds}{dt}\right)\,dt = \int ds$
- Integrating ds gives s, which is displacement.
- Therefore, the area under the VT graph provides the displacement of the object.

Significance of the VT Graph

- The VT graph is highly important because it beautifully captures the three most critical parameters of motion:
  1. **Velocity:** Can be directly read from the y-axis at any given time.
  2. **Acceleration:** Obtained from the slope of the graph.
  3. **Displacement:** Obtained from the area under the graph.

Concavity and Convexity in Position-Time Graphs

Introduction to Concavity and Convexity

In position-time or displacement-time graphs, understanding concavity and convexity provides crucial information about the object's acceleration.

Types of Concavity/Convexity:

- **Concave Up (or Convex Down):**
  - These graphs resemble a 'U' shape opening upwards.
  - Examples: Graph 1 and Graph 2 (as described in the lecture).
  - **Analogy:** "Filling water" – if water were poured onto the curve, it would collect and be stored.
- **Concave Down (or Convex Up):**
  - These graphs resemble an inverted 'U' shape, opening downwards.

Notes

Deducing Acceleration from Concavity:

1. **For Concave Up Graphs:**
   - Observe the slope of the graph. As you move along a concave up curve from left to right, the slope continuously **increases**.
   - In a position-time graph, the slope represents **velocity**.
   - Therefore, if the slope (velocity) is continuously increasing, it implies that the **acceleration is positive**.
   - **Conclusion:** A concave up position-time graph indicates **positive acceleration**.
2. **For Concave Down Graphs:**
   - Observe the slope of the graph. As you move along a concave down curve from left to right, the slope continuously **decreases**.
   - Since the slope represents velocity, a decreasing slope means the **velocity is decreasing**.
   - Therefore, if the velocity is continuously decreasing, it implies that the **acceleration is negative**.
   - **Conclusion:** A concave down position-time graph indicates **negative acceleration**.

Acceleration-Time (AT) Graph

Understanding the AT Graph

In an acceleration-time (AT) graph:

- **Y-axis:** Represents acceleration (A).
- **X-axis:** Represents time (t).

Area Under the AT Graph

The area under an AT graph gives the **change in velocity** (Δv), not velocity itself. This is a common point of confusion and has been tested in exams like JEE.

- **Mathematical Derivation:** The area is given by the integral of $y\,dx$. In this case, $y = A$ and $dx = dt$. We know that acceleration is the rate of change of velocity, so $A = \frac{dv}{dt}$. Therefore, $dv = A\,dt$. Integrating both sides: $\int dv = \int A\,dt$ The integral of $A\,dt$ represents the area under the AT graph. Let the initial velocity be u and the final velocity be v. $\int_u^v dv = [v]_u^v = v - u$ Thus, the area under the AT graph from time t_1 to t_2 is equal to the change in velocity during that interval: Area = $\Delta v = v_{\text{final}} - v_{\text{initial}}$

Slope of the AT Graph

- This quantity, $\frac{dA}{dt}$, is known as **jerk**.
- Jerk represents the rate of change of acceleration.
- **Note:** While it's good to know for general knowledge, the concept of jerk (rate of change of acceleration) is generally **not part of the JEE syllabus** and can be ignored for exam preparation. It describes the sensation felt when acceleration changes suddenly (e.g., pressing the accelerator suddenly in a vehicle).

Understanding Graphs in Physics

Introduction to Graph Interpretation

In physics, various types of graphs can be drawn to represent physical quantities and their relationships. There is no limit to the combinations of quantities that can be plotted. The fundamental principles for interpreting any graph remain consistent:

- **Slope:** The slope of a graph, given by dy/dx, represents the rate of change of the quantity on the y-axis with respect to the quantity on the x-axis.
- **Area Under the Curve:** The area under the curve, given by $\int y\,dx$, represents the product of the quantity on the y-axis and the change in the quantity on the x-axis.

Example 1: Force-Time Graph

Let's consider a force-time graph. To determine what the area under this graph represents, we apply the fundamental rule:

- **Area under Force-Time Graph:** $\int F\,dt$
- We know that force (F) is defined as the rate of change of momentum (p): $F = dp/dt$.
- Substituting this into the integral: $\int (dp/dt)\,dt = \int dp$
- Integrating dp from an initial momentum to a final momentum gives the change in momentum, Δp.
- The change in momentum is also known as **impulse**.

Therefore, the area under a force-time graph gives the **impulse**.

Example 2: Current-Time Graph

Similarly, let's analyze a current-time graph:

- **Area under Current-Time Graph:** $\int I\,dt$
- We know that current (I) is defined as the rate of flow of charge (q): $I = dq/dt$.
- Substituting this into the integral: $\int (dq/dt)\,dt = \int dq$

Notes

Key Takeaway

The ability to interpret graphs by understanding the meaning of their slope and area under the curve is crucial in physics. It allows you to deduce relationships between physical quantities without memorizing every possible graph interpretation. Physics emphasizes thinking and understanding principles rather than rote memorization.

Area Under Graphs: Displacement, Distance, and Change in Velocity

Key Concepts

- **Area under Velocity-Time Graph:** The area under a velocity-time graph represents **displacement**, not distance. Displacement is a vector quantity that refers to the overall change in position of an object.
- **Area under Speed-Time Graph:** To find the **distance** traveled, one must consider the area under a speed-time graph. Distance is a scalar quantity representing the total path length covered.
- **Area under Acceleration-Time Graph:** The area under an acceleration-time graph gives the **change in velocity** of an object. This is also a vector quantity.

Important Distinction

- **Displacement vs. Distance:** It is crucial to understand that displacement and distance are generally not the same. They are only equal when an object moves in a single direction without changing its path.

Summary of Statements

- **Statement 1:** "Area under velocity-time graph gives distance." This statement is **incorrect** because the area under a velocity-time graph gives displacement.
- **Statement 2:** "Area under acceleration-time graph gives change in velocity." This statement is **correct**.

Problem: Position from V-T Graph

Question

A particle starts from the origin at time $t = 0$ and moves along the positive x-axis. The graph of velocity with respect to time (V-T graph) is given. What is the position of the particle at time $t = 5$ seconds?

Understanding the Problem

- A velocity-time (V-T) graph is provided.
- We need to find the particle's position at $t = 5$ seconds.

Solution Approach

To find the position of the particle, we need to determine its displacement. For a V-T graph, the displacement of a particle is given by the area under the velocity-time curve.

Step-by-Step Calculation

1. **Identify the relevant time interval:** We need to find the position at $t = 5$ seconds, so we will calculate the area under the V-T graph from $t = 0$ to $t = 5$ seconds.
2. **Calculate the area:** By visually inspecting the graph (which is assumed to be a grid of unit squares, where each square represents a certain displacement):
   - Count the full squares under the curve from $t = 0$ to $t = 5$.
   - Count the partial squares and combine them to form full squares.

   Let's assume the graph shows:

   - 8 full squares.
   - Two half-squares, which combine to form 1 full square.

   Total area = 8 (full squares) + 1 (from two half-squares) = 9 units.

   Since the y-axis represents velocity (e.g., m/s) and the x-axis represents time (s), the area units will be meters (m), representing displacement.

   So, the displacement of the particle from $t = 0$ to $t = 5$ seconds is 9 meters.
3. **Determine the final position:**
   - The particle starts from the origin, meaning its initial position (x_0) is 0 meters.
   - Displacement (Δx) is defined as the change in position: $\Delta x = x_f - x_0$.
   - We found $\Delta x = 9$ meters.
   - Therefore, $9 = x_f - 0$.
   - $x_f = 9$ meters.

Conclusion

The position of the particle at $t = 5$ seconds is 9 meters along the positive x-axis.

This type of question is a common JEE (Joint Entrance Examination) previous year's question (PYQ).

Notes

This section discusses how to analyze the motion of a particle moving with constant acceleration, specifically focusing on its representation through acceleration-time (AT), velocity-time (VT), and position-time (XT) graphs.

Scenario

Imagine a particle (like a cockroach) starting from the origin and moving along the positive x-axis with a constant positive acceleration. This means the acceleration is uniform and directed in the positive x-direction.

Acceleration-Time (AT) Graph

- Since the acceleration is constant, its value does not change with time.
- Therefore, the AT graph will be a horizontal line. This indicates that acceleration remains the same at all points in time.

Velocity-Time (VT) Graph

- The slope of the VT graph represents acceleration.
- Given that acceleration is constant, the slope of the VT graph must also be constant.
- A graph with a constant slope is a straight line.
- Since the particle starts from rest (zero velocity) at the origin, the VT graph will be a straight line passing through the origin.

Position-Time (XT) Graph

- The XT graph represents the position of the particle over time.
- For motion with constant positive acceleration, the XT graph will be a parabola that is concave up.
- **Concave Up:** This shape signifies positive acceleration. Think of it as a "filling water" shape.
- **Concave Down:** In contrast, a concave down shape would indicate negative acceleration (a "spilling water" shape).

Conclusion

When a particle moves with constant positive acceleration starting from rest at the origin:

- The AT graph is a horizontal line.
- The VT graph is a straight line passing through the origin.
- The XT graph is a parabola that is concave up.

Notes



Problem Statement

The position-time graphs for two students, A and B, returning from school to their homes are shown in the figure. We need to analyze the graph and determine the correct statements among the given options:

(A) A lives closer to the school (B) B lives closer to the school (C) A takes lesser time to reach home (D) A travels faster than B (E) B travels faster than A

We then choose the correct combination of these statements.

Graph Interpretation

Let's consider the school as the origin ($x = 0$). Both students are moving away from the school towards their homes.

1. **Distance to Home (Position at which they stop):**
   - Student A starts at $T = 0$ and their position graph ends at a certain point, let's say $X_A = 4$ units. This indicates that A's home is at $X = 4$.
   - Student B starts later (e.g., at $T = 1$ second) and their position graph ends at a point further away, let's say $X_B = 6$ units. This indicates that B's home is at $X = 6$.
   - Since $X_B > X_A$, B's home is farther from the school than A's home. Therefore, **statement (A) "A lives closer to the school" is correct**, and statement (B) "B lives closer to the school" is incorrect.
2. **Time Taken to Reach Home:**
   - Student A starts at $T = 0$ and reaches home (stops) at approximately $T = 5$ seconds. So, A takes $5 - 0 = 5$ seconds to reach home.
   - Student B starts at approximately $T = 1$ second and also reaches home (stops) at approximately $T = 5$ seconds. So, B takes $5 - 1 = 4$ seconds to reach home.
   - Since B takes 4 seconds and A takes 5 seconds, B takes less time to reach home. Therefore, **statement (C) "A takes lesser time to reach home" is incorrect**.
3. **Speed/Velocity Comparison:**

- Alternatively, let's compare distance and time:
  - A travels a distance of 4 units in 5 seconds.
  - B travels a distance of 6 units (from $X = 0$ to $X = 6$) in 4 seconds (from $T = 1$ to $T = 5$).
  - Since B covers a greater distance in less time, B is traveling faster than A. Therefore, **statement (D) "A travels faster than B" is incorrect**, and **statement (E) "B travels faster than A" is correct**.

Conclusion

Based on the analysis:

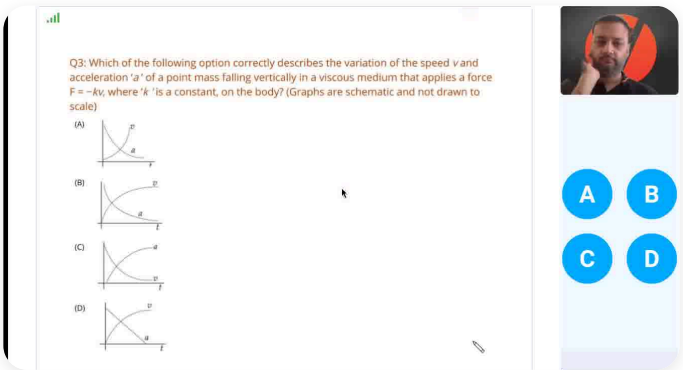
- Statement (A) is correct.
- Statement (E) is correct.

Thus, the correct option is the one that includes (A) and (E).

Final Answer

The correct answer is (B) (A) and (E) only.

Problem: Particle Falling in a Viscous Medium



Question

Which of the following options correctly describes the variation of the speed v and acceleration 'a' of a point mass falling vertically in a viscous medium that applies a force $F = -kv$, where 'k' is a constant, on the body? (Graphs are schematic and not drawn to scale)

(A)

(B)

(C)

Notes

When a particle falls in a viscous medium, it experiences two primary forces:

1. **Gravitational Force (mg):** Acting downwards.
2. **Viscous Force (kv):** Acting upwards, opposing the motion. This force is proportional to the velocity (v) of the particle, with 'k' being the constant of proportionality.

Net Force and Acceleration

Taking the downward direction as positive, the net force (F_{net}) acting on the particle is: $F_{net} = mg - kv$ According to Newton's second law, $F_{net} = ma$. Therefore, the acceleration (a) of the particle is: $ma = mg - kv$ $a = \frac{mg-kv}{m}$ $a = g - \frac{kv}{m}$

Analysis of Acceleration's Variation

- As the particle starts falling, its velocity (v) increases.
- Since $a = g - \frac{kv}{m}$, as v increases, the term $\frac{kv}{m}$ increases.
- Consequently, the acceleration (a) decreases. This means the rate at which velocity increases slows down over time.
- This behavior is characteristic of reaching a terminal velocity, where acceleration eventually becomes zero.

Eliminating Incorrect Options Based on Acceleration

- Graphs where acceleration is shown to be increasing are incorrect. Based on our analysis, acceleration must decrease with time.

Derivation of Velocity and Acceleration Functions

Velocity as a Function of Time

We know that acceleration $a = \frac{dv}{dt}$. So, we can write: $\frac{dv}{dt} = g - \frac{kv}{m}$ To solve this differential equation, we separate variables: $\frac{dv}{g-\frac{kv}{m}} = dt$ Integrate both sides. Let $u = g - \frac{kv}{m}$, then $du = -\frac{k}{m}dv$, so $dv = -\frac{m}{k}du$. $\int_0^v \frac{dv}{g-\frac{kv}{m}}$ $\ln(g - \frac{kv}{m}) - \ln(g)] = t$ $\ln\left(\frac{g-\frac{kv}{m}}{g}\right) = -\frac{kt}{m}$ $\frac{g-\frac{kv}{m}}{g} = e^{-\frac{kt}{m}}$ $1 - \frac{kv}{mg} = e^{-\frac{kt}{m}}$ $\frac{kv}{mg} = 1 - e^{-\frac{kt}{m}}$ $v(t) = \frac{mg}{k}(1 - e^{-\frac{kt}{m}})$

Did you find these notes helpful?

Analysis of Velocity Function

- At $t = 0$: $v(0) = \frac{mg}{k}(1 - e^0) = \frac{mg}{k}(1 - 1) = 0$. This is consistent with the particle being released from rest.
- As $t \to \infty$: $e^{-\frac{kt}{m}} \to 0$. Therefore, $v(\infty) = \frac{mg}{k}(1 - 0) = \frac{mg}{k}$. This constant velocity is known as the terminal velocity.

Notes

Acceleration as a Function of Time

Substitute the expression for $v(t)$ back into the acceleration equation $a = g - \frac{kv}{m}$: $a(t) = g - \frac{k}{m}\left[\frac{mg}{k}(1 - e^{-\frac{kt}{m}})\right]$ $a(t) = g - g(1 - e^{-\frac{kt}{m}})$ $a(t) = g - g + ge^{-\frac{kt}{m}}$ $a(t) = ge^{-\frac{kt}{m}}$

Analysis of Acceleration Function

- At $t = 0$: $a(0) = ge^0 = g$. This is the initial acceleration due to gravity.
- As $t \to \infty$: $e^{-\frac{kt}{m}} \to 0$. Therefore, $a(\infty) = g \times 0 = 0$. This is consistent with reaching terminal velocity where acceleration becomes zero.
- The acceleration decreases non-linearly (exponentially) from g to 0. This means the acceleration-time graph should be a decaying exponential curve, not a straight line.

Conclusion for the Problem

Based on the derived functions:

- Velocity increases non-linearly and approaches a constant value (terminal velocity).
- Acceleration decreases non-linearly (exponentially) and approaches zero.

Comparing these findings with the given options:

- Option D shows acceleration decreasing linearly, which is incorrect.
- Option B correctly depicts velocity increasing non-linearly and flattening out (approaching terminal velocity), and acceleration decreasing non-linearly (exponentially) towards zero.

Therefore, option B is the correct answer.

Understanding Acceleration in Projectile Motion

Key Concept: Acceleration Due to Gravity

When an object is thrown into the air, its acceleration remains constant throughout its flight (ignoring air resistance). This constant acceleration is due to gravity, which always acts downwards. The magnitude of this acceleration is approximately $9.8\ m/s^2$ (or g).

Velocity-Time (v-t) Graphs and Acceleration

- The slope of a velocity-time (v-t) graph represents acceleration.
- Since the acceleration of an object in projectile motion (after being thrown) is constant, the slope of its v-t graph must also be constant.

Analyzing v-t Graph Options

versa, it implies that the acceleration is changing. This is incorrect for an object under gravity.

- **Constant Slope:** A v-t graph with a constant slope indicates constant acceleration. This is the correct representation for an object thrown into the air.

Example: Object Thrown Upwards

When an object is thrown upwards, its initial velocity is positive (if upward is taken as positive). As it moves upwards, gravity acts downwards, causing the velocity to decrease. At its peak, the velocity momentarily becomes zero. As it falls back down, its velocity becomes negative and increases in magnitude. Throughout this entire motion, the acceleration remains constant and directed downwards.

Therefore, the v-t graph for such motion should be a straight line with a constant negative slope (if upward is positive, or a constant positive slope if downward is positive).

Problem: Velocity-Time Graph Analysis

Question

The velocity-time graph of a car and a scooter are shown. We need to find two things:

1. The difference between the distances traveled by the car and the scooter in 15 seconds.
2. The time at which the car will catch up with the scooter.

Given Information (from graph, not explicitly provided in text but implied):

- **Scooter:** Moves with a constant velocity of $30\ \text{m/s}$.
- **Car:** Starts from rest and accelerates, reaching a velocity of $45\ \text{m/s}$ at $t = 15\ \text{s}$. The velocity-time graph for the car is a straight line from $(0, 0)$ to $(15, 45)$.

Distance Traveled in 15 Seconds

Distance Traveled by Scooter in 15 seconds

Since the scooter moves with a constant velocity, its displacement is given by:
$\text{Displacement}_{\text{scooter}} = \text{Velocity} \times \text{Time}$ $\text{Displacement}_{\text{scooter}} = 30\ \text{m/s} \times 15\ \text{s} = 450\ \text{m}$

Distance Traveled by Car in 15 seconds

The displacement of the car is the area under its velocity-time graph. Since the car's graph is a triangle up to 15 seconds, the area is: $\text{Displacement}_{\text{car}} = \frac{1}{2} \times \text{Base} \times \text{Height}$
$\text{Displacement}_{\text{car}} = \frac{1}{2} \times 15\ \text{s} \times 45\ \text{m/s}$ $\text{Displacement}_{\text{car}} = \frac{1}{2} \times 675 = 337.5\ \text{m}$

Difference in Distances at 15 seconds

Notes

Time When Car Catches Up with Scooter

We need to find the time (t) at which the displacement of the car equals the displacement of the scooter. Let's analyze the options or calculate directly.

Analyzing the situation after 15 seconds

At $t = 15$ s, the car is still behind the scooter (112.5 m behind). This means the car will catch up with the scooter at a time greater than 15 s.

Let's consider the time $t = 22.5$ s.

Displacement of Scooter at $t = 22.5$ s

- From 0 to 15 s: 450 m
- From 15 to 22.5 s: The scooter continues at 30 m/s for an additional $22.5 - 15 = 7.5$ s. $\text{Displacement}_{\text{scooter, 15-22.5s}} = 30 \text{ m/s} \times 7.5 \text{ s} = 225 \text{ m}$
- Total displacement of scooter at 22.5 s: $\text{Total Displacement}_{\text{scooter}} = 450 \text{ m} + 225 \text{ m} = 675 \text{ m}$

Displacement of Car at $t = 22.5$ s

- From 0 to 15 s: 337.5 m
- From 15 to 22.5 s: The car's velocity at $t = 15$ s is 45 m/s. Assuming the car maintains this velocity (or continues its acceleration as per the graph, which implies constant velocity after 15s if the graph flattens, or continues to accelerate if it's a straight line passing through (15,45) and beyond). Based on the problem context, it seems the car reaches 45 m/s at 15 s and then maintains that velocity, or the graph continues to be a straight line. If the graph is a straight line from $(0, 0)$ to $(15, 45)$ and then continues horizontally, the car's velocity is 45 m/s from 15 s onwards. The additional displacement would be: $\text{Displacement}_{\text{car, 15-22.5s}} = 45 \text{ m/s} \times (22.5 - 15) \text{ s} = 45 \text{ m/s} \times 7.5 \text{ s} = 337.5 \text{ m}$
- Total displacement of car at 22.5 s: $\text{Total Displacement}_{\text{car}} = 337.5 \text{ m} + 337.5 \text{ m} = 675 \text{ m}$

Conclusion

At $t = 22.5$ s, both the car and the scooter have traveled a total distance of 675 m. Therefore, the car catches up with the scooter at 22.5 s.

Analyzing Velocity vs. Height Graph for a Falling Ball

Problem Statement

Notes

Initial Conditions

- **Release Point:** Height H
- **Initial Velocity:** $v = 0$ at height H.

Motion Analysis (Falling Downwards)

1. **Velocity at Height H:** At the initial height H, the velocity of the ball is 0. This means the graph must start at $(H, 0)$.
2. **Velocity Direction:** As the ball falls, its velocity will be in the negative direction (downwards). Therefore, the velocity values will be negative.
3. **Speed Increase:** As the ball falls, its speed increases. This means the magnitude of the negative velocity increases.
4. **Relationship between Velocity and Height:**
   - The kinematic equation relating initial velocity (u), final velocity (v), acceleration (g), and displacement (s) is $v^2 = u^2 + 2as$.
   - In this case, $u = 0$, $a = -g$ (if we consider upward as positive), and displacement is $(h - H)$. So, $v^2 = 0^2 + 2(-g)(h - H)$.
   - This simplifies to $v^2 = -2g(h - H)$ or $v^2 = 2g(H - h)$.
   - This equation shows that v^2 is linearly dependent on height h. However, v itself is not linearly dependent on h. The graph of v versus h will be parabolic, not linear. This eliminates options showing linear relationships.
   - Specifically, as h decreases from H to 0, $H - h$ increases, and thus $|v|$ increases. Since the ball is moving downwards, v is negative.

Motion Analysis (Rebounding Upwards)

1. **Impact and Rebound:** When the ball hits the floor (height $h = 0$), it rebounds. The problem states it reaches a height $H/2$.
2. **Velocity at Maximum Rebound Height:** At the maximum rebound height $H/2$, the ball momentarily stops, meaning its velocity becomes 0.
3. **Velocity Direction (Rebound):** As the ball moves upwards after the bounce, its velocity will be in the positive direction.
4. **Speed Decrease (Rebound):** As the ball moves upwards, its speed decreases until it reaches 0 at $H/2$.

Evaluating Graph Characteristics

- **Initial State:** The graph must show velocity 0 at height H.

change in velocity, and then it becomes faster as the ball gains speed.

- **Rebound Phase:** After hitting the ground, the ball's velocity becomes positive and decreases as it moves up to $H/2$, where the velocity becomes 0. The curve for this phase should also be parabolic.

Conclusion

Graph A correctly depicts these characteristics:

- Starts at $(H, 0)$.
- Shows a parabolic curve where velocity becomes negative and its magnitude increases as height decreases during the fall.
- Shows velocity becoming 0 again at height $H/2$ after the rebound, with positive velocity during the upward motion.
- The curvature correctly represents that the ball travels slowly initially (height decreases slowly) and then faster (height decreases rapidly) as it falls, and vice-versa during the rebound.

Homework and Doubt Clarification

Homework Questions

The lecturer assigned several questions as homework, encouraging students to attempt them. These questions are primarily from previous year's question papers (PYQs).

Clarification on Velocity and Acceleration

- **dv/dh vs. dv/dt:** A student asked if "dv/dh" is equal to "-g". The lecturer clarified that this is incorrect. The correct relationship is that the rate of change of velocity with respect to time, "dv/dt", is equal to "-g" (acceleration due to gravity).
  - $\frac{dv}{dt} = -g$
- **Position vs. Displacement:** The lecturer explained that while change in position is displacement, both the rate of change of position with respect to time ($\frac{dx}{dt}$) and the rate of change of displacement with respect to time ($\frac{ds}{dt}$) give velocity. For small changes, dx and ds are considered equal.
  - $v = \frac{dx}{dt} = \frac{ds}{dt}$

Uniform Acceleration

- **Definition:** Uniform acceleration means that the acceleration is constant.

Graph Interpretation (VT Graph)

- **Assumptions in Problems:** If a problem does not explicitly mention conditions like a ball hitting the ground and bouncing back, such assumptions should not be made. The graph typically represents the velocity variation with time when an object is thrown upwards under gravity, assuming standard conventions (e.g., upward as positive, downward as negative).

Motion along X-axis

- If a particle starts from rest and moves along the positive x-axis with acceleration, the acceleration must be positive for it to continue moving in the positive direction.

Velocity and Height Relationship

- Velocity increases at a constant rate with respect to time under uniform acceleration. However, with respect to height, velocity increases more rapidly as height decreases slowly.

Study Advice

- **Problem Solving:** The lecturer emphasized that solving problems is crucial for success, more so than just reading books. Students are advised to solve a lot of problems, including redoing the questions covered in class and the provided JPYQs.
- **Resources:** A VDPP (Vedantu Daily Practice Problems) with respect to graphs will be shared. Homework problems will also be provided in PDF format via the Telegram group.
- **Time Management:** Students are encouraged to make a timetable and manage their time effectively, especially since school-related concerns might be less pressing.
- **Physics Study:** For physics, solving problems and practicing with provided materials is sufficient; additional external resources are not necessarily required.